

CSES Dynamic Programming — Problem Reference

All listed tasks: 1 s, 512 MB. Examples omitted for compactness.

1093 — Two Sets II

Count the ways to partition $\{1, 2, \dots, n\}$ into two sets having equal sum. The two sets are unordered: swapping the two sides is the same partition. **In:** n

Out: number of partitions modulo $10^9 + 7$

Constraints: $1 \leq n \leq 500$.

1097 — Removal Game

Two players alternately remove the first or last number of a list and add it to their score. Both play optimally. Find the maximum final score the first player can guarantee. **In:** n ; then $x_1 \dots x_n$

Out: maximum score of the first player

Constraints: $1 \leq n \leq 5000$, $-10^9 \leq x_i \leq 10^9$. Use 64-bit.

1140 — Projects

Project i occupies every day from a_i through b_i and gives reward p_i . Choose non-overlapping projects, with at most one attended on any day, maximizing total reward. **In:** n ; then $a_1 \dots b_1 \dots p_1$

Out: maximum total reward

Constraints: $n \leq 2 \cdot 10^5$, $1 \leq a_i \leq b_i \leq 10^9$, $p_i \leq 10^9$. Use 64-bit.

1145 — Increasing Subsequence

Given an integer array, find the length of its longest *strictly increasing* subsequence. A subsequence preserves order but may skip elements. **In:** n ; then $x_1 \dots x_n$

Out: LIS length

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

1158 — Book Shop

There are n books; book i costs h_i and has s_i pages. Buy each book at most once, spending at most x , and maximize total pages. **In:** n x ; prices h_i ; pages s_i

Out: maximum number of pages

Constraints: $n \leq 1000$, $x \leq 10^5$, $1 \leq h_i, s_i \leq 1000$.

1633 — Dice Combinations

Count ordered sequences of dice throws from $\{1, 2, \dots, 6\}$ whose sum is exactly n . **In:** n

Out: number of sequences modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10^6$.

1634 — Minimizing Coins

Given distinct positive coin values, each usable any number of times, form sum x using the minimum number of coins. **In:** n x ; then $c_1 \dots c_n$

Out: minimum coin count, or -1 if impossible

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1635 — Coin Combinations I

Given distinct positive coin values, each reusable, count *ordered* sequences of coins whose values sum to x . Different orders count separately. **In:** n x ; then $c_1 \dots c_n$

Out: number of ordered combinations modulo $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1636 — Coin Combinations II

Given distinct positive coin values, each reusable, count combinations whose values sum to x when order does *not* matter. **In:** n x ; then $c_1 \dots c_n$

Out: number of combinations modulo $10^9 + 7$

Constraints: $n \leq 100$, $x \leq 10^6$, $1 \leq c_i \leq 10^6$.

1637 — Removing Digits

Starting from n , each step subtracts one decimal digit currently appearing in the number. Find the minimum steps needed to reach 0. **In:** n

Out: minimum number of steps

Constraints: $1 \leq n \leq 10^6$.

1638 — Grid Paths I

An $n \times n$ grid contains empty cells $.$ and traps $*$. Move from the upper-left to the lower-right using only right/down moves and never enter a trap. **In:** n ; then n grid rows

Out: number of valid paths modulo $10^9 + 7$

Constraints: $1 \leq n \leq 1000$.

1639 — Edit Distance

Find the minimum number of single-character insertions, deletions, and replacements needed to transform one uppercase string into the other. **In:** first string; second string

Out: edit distance

Constraints: $1 \leq n, m \leq 5000$.

1653 — Elevator Rides

There are n people with weights w_i and an elevator capacity x . Every person must reach the top; each ride may carry any subset whose total weight is at most x . Minimize rides. **In:** n x ; then $w_1 \dots w_n$

Out: minimum number of rides

Constraints: $1 \leq n \leq 20$, $1 \leq w_i \leq x \leq 10^9$.

1744 — Rectangle Cutting

Cut an $a \times b$ integer-sided rectangle into squares. Each move cuts one current rectangle into two rectangles along an integer coordinate. Minimize the number of cuts. **In:** a b

Out: minimum number of cuts

Constraints: $1 \leq a, b \leq 500$.

1745 — Money Sums

Given n positive-value coins, each usable at most once, find every positive sum obtainable from a subset. **In:** n ; then $x_1 \dots x_n$

Out: number k of distinct positive sums; then all sums in increasing order

Constraints: $1 \leq n \leq 100$, $1 \leq x_i \leq 1000$.

1746 — Array Description

Count arrays of length n with values in $[1, m]$ such that adjacent values differ by at most 1. Input positions may be fixed; value 0 means unknown. **In:** n m ; then $x_1 \dots x_n$

Out: number of matching arrays modulo $10^9 + 7$

Constraints: $n \leq 10^5$, $m \leq 100$, $0 \leq x_i \leq m$.

1748 — Increasing Subsequence II

Count all nonempty *strictly increasing* subsequences of an integer array. Subsequences using different positions are distinct even if their values match. **In:** n ; then $x_1 \dots x_n$

Out: number of increasing subsequences modulo $10^9 + 7$

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq x_i \leq 10^9$.

2181 — Counting Tilings

Count ways to tile an $n \times m$ rectangle completely using 1×2 and 2×1 dominoes. **In:** n m

Out: number of tilings modulo $10^9 + 7$

Constraints: $1 \leq n \leq 10$, $1 \leq m \leq 1000$.

2220 — Counting Numbers

Count integers in the inclusive range $[a, b]$ whose decimal representation has no two equal adjacent digits. **In:** a b

Out: number of valid integers

Constraints: $0 \leq a \leq b \leq 10^{18}$. Use 64-bit.

2413 — Counting Towers

For each query n , count different towers of width 2 and height n built from integer-sided blocks. Distinct-looking mirrored or rotated constructions count separately. **In:** t ; then one n per test

Out: answer for each test modulo $10^9 + 7$

Constraints: $1 \leq t \leq 100$, $1 \leq n \leq 10^6$.

3314 — Mountain Range

Mountains stand in a row. You may glide from mountain a to b when a is taller than b and taller than every mountain strictly between them. Start anywhere. Maximize mountains visited. **In:** n ; then $h_1 \dots h_n$

Out: maximum number of visited mountains

Constraints: $1 \leq n \leq 2 \cdot 10^5$, $1 \leq h_i \leq 10^9$.

3359 — Minimal Grid Path

An $n \times n$ grid contains uppercase letters. Move from the upper-left to the lower-right using only right/down moves; concatenate every visited cell. Find the lexicographically smallest possible string. **In:** n ; then n rows of n letters

Out: lexicographically minimum path string

Constraints: $1 \leq n \leq 3000$.

3403 — Longest Common Subsequence

Given two integer arrays, find a longest sequence obtainable as a subsequence of both. A subsequence preserves order but may skip elements. **In:** n m ; array a ; array b

Out: LCS length; then any one longest common subsequence

Constraints: $1 \leq n, m \leq 1000$, $1 \leq a_i, b_i \leq 10^9$.